

Open System-One Decision Models: Frozen Diffusion-LM Backbones and Specialist Decision Heads for Calibrated Scientific Decisions

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Abstract

Structured decisions—selecting among options, verifying whether evidence supports a claim, and grading response quality on ordinal scales—are the atomic operations of automated scientific pipelines. Proprietary “System-One” models deliver such decisions as typed API calls without generation, but are closed. We present an open, trainable counterpart: a frozen masked-diffusion language model backbone (LLaDA-8B) with small specialist decision heads producing probabilities for three decision types (choice, noul, score). We contribute three findings. (1) A deterministic canonical ordering of options yields exact permutation invariance (option-order flip rate $0.77 \rightarrow 0.00$), where train-time logit averaging does not. (2) Multi-layer attention pooling over option spans outperforms last-layer mean pooling for choice (+7 points), while an ordinal auxiliary loss improves graded scoring (+9 points). (3) Scaling training data $30\times$ on an out-of-domain corpus improves in-distribution accuracy to 0.96 but does not transfer (−13 points on the target domain), showing that coverage of the target reasoning distribution—not volume—is the binding constraint. On public benchmarks, OURS reaches 0.853 on SciFact claim verification and transfers zero-shot to general classification (0.930 SST-2, 0.854 AG News), while parametric knowledge tasks remain bounded by the backbone ceiling (0.315 GPQA vs. backbone ≈ 0.33).

1 Introduction

Automated scientific pipelines—agentic literature review, claim verification, data extraction, routing to slower reasoning components—are built out of small structured judgments: *which* of these options answers the question, *does* the evidence support the claim, *how* good is this candidate answer. Today these judgments are typically extracted from generative LLMs by prompting for JSON and parsing the output, a pipeline that is slow, brittle under format drift, and poorly calibrated.

A recent class of models removes generation entirely: given an unstructured state and a set of typed questions, a *System-One* model returns typed values—a choice among options, a yes/no probability, an ordinal score—each with a calibrated probability, in a single forward pass. The flagship proprietary model of this class is Jev (TypeSafe AI, 2026); open reimplementations appeared within days of its release (kev, openjev, laya, SemIf), all built on autoregressive or encoder backbones.

We ask what an open System-One model needs to be good at, and build one on the other major family of language models: a *masked-diffusion* backbone (LLaDA-8B), whose bidirectional attention is a natural fit for decision readouts that compare each option against the full context. Rather than fine-tuning the backbone—which we show causes catastrophic forgetting—we train small specialist heads per decision type on frozen multi-layer hidden states. Our contributions:

- **Exact permutation invariance by canonical ordering.** Sorting options deterministically by content makes the presentation order a pure function of the option set: the option-order flip rate is 0.00 by construction, where train-time logit averaging neither achieves invariance nor preserves accuracy.
- **Specialist heads per decision type.** Multi-layer attention pooling over option spans improves choice (+7 points over mean pooling) while hurting binary verification—different decision types want different readouts. An ordinal auxiliary loss improves graded scoring (+9 points).
- **The backbone is the knowledge bound.** A $30\times$ out-of-domain data increase fails to transfer (-13 points), and on GPQA the head reaches the backbone’s reported ceiling (0.315 vs. ≈ 0.33): decision accuracy on knowledge tasks is a property of the backbone, not the readout. Domain knowledge must be injected into the weights—which we do via LoRA domain-adaptive pretraining on scientific full texts.

We evaluate on a passage-grounded scientific decision battery and on public benchmarks (GPQA, SciFact, and four general classification sets), reporting accuracy, calibration, option-order flip rate, and automation at 5% error.

2 Setup

Decision types. We consider three typed decisions over a state (context) and a question: **choice** selects one of K answer options; **noul** decides whether evidence supports a claim ($K=2$); **score** assigns an ordinal grade $0 < 1 < 2$ (unrelated / related-but-wrong / correct).

Model. Each example is encoded as a single masked sequence: context, question, and options separated by marker tokens. A frozen masked-diffusion backbone (LLaDA-8B-Instruct) produces token representations; a specialist head per decision type maps pooled option-span features to class logits. Two head architectures are compared: a mean-pooled MLP and *AttnPool*, a learned query attending over each option span on concatenated hidden states from layers $\{8, 16, 24, 32\}$. Only the heads are trained ($\sim 2\text{M}$ parameters); the backbone remains frozen.

Canonical ordering. Options are sorted by token-id content before encoding, and gold indices remapped. The presentation order is thus a pure function of the option *set*: any input permutation collapses to the same sequence and permutation invariance holds exactly, at training, calibration, and evaluation.

Ordinal auxiliary loss. For score we add a CORAL-style term $\sum_{j=1}^{K-1} \text{BCE}(P(y \geq j), \mathbf{1}[g \geq j])$ on cumulative softmax mass, respecting the grade order without changing the K -way decode.

Metrics. Accuracy, expected calibration error (ECE), option-order flip rate (accuracy change under input permutation), and *automation@5%err*: the fraction of decisions that can be auto-accepted while keeping realized error $\leq 5\%$ —the operating metric for a decision layer.

3 Experiments

3.1 Scientific decision battery

The *elite* battery is built from passage-grounded scientific QA typed by reasoning kind (factual, causal, multihop, negation, numerical, definitional); each passage generates a choice item ($K=4$, one gold + three distractors), a noul item (supported vs. unsupported claim), and a score item (correct / related-but-wrong / unrelated answer). Splits are passage-disjoint to prevent leakage (2,240/6,720/6,720 train rows per type). The *v2* battery scales the same construction to 25,000 passages from three raw QA corpora (physics/polymer LaTeX full texts, astronomy, and a derived set) with groundedness filters—content-word recall ≥ 0.5 and numeric binding—yielding $\sim 470,000$ training decisions; we use it to probe the data-scaling question.

3.2 Public benchmarks

GPQA main (448 graduate-level questions) and diamond subset (198) as choice; SciFact dev (340 claim–abstract pairs) as noul and 3-way score; plus general classification (SST-2, AG News, Enron spam, Banking77) to probe out-of-domain transfer. Benchmarks are eval-only; temperature is fitted on the battery dev split (no benchmark leakage).

4 Results

4.1 Ablations on the scientific battery

	choice	noul	score	flip (choice)
Baseline (mixed MLP)	0.775	0.707	0.522	0.76
+ orders= 4	0.721	0.782	0.586	–
+ AttnPool	0.847	0.636	0.588	0.75
+ both	0.749	0.780	0.576	0.70
Per-type heads	0.876	0.710	0.574	0.77
Per-type + canon. + ord.	0.870	0.783	0.608	0.00

Table 1: Ablation on the scientific battery (frozen LLaDA-8B). Canonical ordering eliminates order sensitivity exactly; per-type specialist heads plus AttnPool give the best choice accuracy and automation.

Key findings: (1) train-time logit averaging over permutations does *not* fix order sensitivity and costs choice accuracy; (2) canonical ordering achieves flip rate exactly 0 at no accuracy cost; (3) AttnPool on intermediate layers helps choice (+7) but hurts noul, motivating per-type heads; (4) the ordinal auxiliary loss lifts score $0.522 \rightarrow 0.608$.

4.2 Data scaling does not transfer across domains

4.3 Public benchmarks

5 Discussion

What a decision layer buys. The value of a System-One component is not only accuracy but *operating characteristics*: determinism (canonical ordering makes predictions a pure function of

	in-distribution (v2)	elite (target)
choice	0.965	0.738
noul	0.942	0.782
score	–	0.616

Table 2: Heads retrained on $30\times$ more data from a different QA distribution (qa_raw corpora) reach near-ceiling in-distribution accuracy but do not transfer to the elite battery, whose questions stress causal/multihop/negation reasoning absent from the source corpus.

benchmark	type	Ours	reference
GPQA main	choice	0.315	GPT-4 ~ 0.39 ; expert ~ 0.65 ; backbone ceiling ~ 0.33
GPQA diamond	choice	0.328	–
SciFact	noul	0.853	–
SciFact	score	0.624	–
Enron spam	choice	0.752	published System-One: 0.987
SST-2	choice	0.930	0.957
AG News	choice	0.854	0.913
Banking77 ($K=77$)	choice	0.229	0.760

Table 3: Zero-shot transfer to public benchmarks (eval-only; temperature fitted on the battery dev split, no benchmark data seen). Published System-One numbers are third-party measurements (Kumar, 2026) on identical public datasets. GPQA accuracy sits at the reported backbone ceiling (LLaDA-8B ≈ 0.33), indicating the readout already extracts the backbone’s accessible parametric knowledge. Transfer to $K \leq 4$ classification is strong (SST-2, AG News); $K=77$ degrades because training used $K \leq 4$.

the option set), calibrated probabilities (ECE 0.03–0.12 after temperature fitting), and a usable automation budget—82% of choice traffic on our scientific battery can be auto-accepted at $\leq 5\%$ realized error, which is what makes routing to a slower System-Two component a quantified decision rather than a heuristic.

Division of labor between backbone and head. Our results consistently point to a clean separation: the readout determines *how well accessible information is used* (pooling, ordering, ordinality), while the backbone determines *what is knowable at all* (GPQA at backbone ceiling; zero-shot transfer of grounding skills to SST-2/AG News; collapse on $K=77$ for lack of training coverage). This suggests the development loop for open decision models: keep the head recipe fixed, improve the backbone’s domain knowledge (domain-adaptive pretraining, or a natively trained domain backbone), and treat benchmarks like GPQA as backbone probes.

Relation to the ecosystem. Concurrent open reimplementations (kev, openjev, laya) share the API contract but differ in backbone and readout; none is diffusion-based, and none reports exact permutation invariance—our canonical ordering achieves it by construction rather than by augmentation. Third-party measurements of the proprietary reference on public datasets (Kumar, 2026) place it well above all open entries on general classification; on scientific verification the margin is much smaller (SciFact noul: ours 0.853 vs. reported ~ 0.89).

6 Limitations

Backbone pretraining may include benchmark-adjacent text (the standard public-benchmark caveat). Heads are trained on one decision distribution; $K > 4$ degrades (option tournamenting is future work), and cross-domain transfer is limited by construction. Score remains the weakest decision type for all measured systems, including the proprietary reference. Our proprietary-reference comparisons cite third-party reports rather than our own measurements.

References

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